Nikhil Kamath x Nandan Nilekani | People by WTF | Ep #3

[Speaker 2] (0:31 - 1:30)

Are we good? Ready guys, should we start? You've done so many interviews now.

No, I feel so intimidated. There is so much information around the India Stack that is so relevant to a young person who wants to be an entrepreneur in India. Hopefully by the end of it, we can learn from you what should be built on top of the India Stack and what is the opportunity today from the lens of capitalism.

Hi Nandan, thank you for doing this. My pleasure Nikhil. I don't think many people know but I end up meeting Nandan most often in his house or when he's travelling to one of hundred different countries that he goes to every year.

That's right. So where are you going next Nandan in your travels?

[Speaker 1] (1:30 - 1:43)

Actually this month is a slow month in the sense I'm going to be right here in Bangalore and in India but I'm going next month to Basel, I'm going to London.

[Speaker 2] (1:43 - 1:48)

All around this technology broadly that you're involved with?

[Speaker 1] (1:49 - 2:21)

Yeah, I basically have two hats. I have a public hat which is really the bulk of what I do which is enabling digital public infrastructure. You know, Aadhaar, UPI, that kind of stuff.

And now we are trying to make that go global. So part of this is the globalization of this DPI concept. And then I have a private sector hat as the chairman of Infosys and a few investments.

So some of my travel is that or if I go on a vacation I just work for a couple of days.

[Speaker 2] (2:21 - 2:26)

I watched a bunch of your interviews. Why would I do that? Over the last 48 hours.

[Speaker 1] (2:26 - 2:28)

You seem to be a sucker for punishment.

[Speaker 2] (2:29 - 3:36)

I feel like I know Nandan, the person in private. I have never heard you speak at length publicly in that way. I've heard you speak at your house when you host dinners and all of that.

So it was actually intriguing and there is so much information around the India stack that is so relevant to a young person who wants to be an entrepreneur in India. So we thought we'll focus today on that. So I'm going to play the part of a 20-year-old who knows nothing about the India stack.

I'll start the journey from the very beginning. And hopefully by the end of it we can learn from you what should be built on top of the India stack and what is the opportunity today from the lens of capitalism. What can me, a 20-year-old start to make money out of it on top of the India stack in any way?

If we were to begin with Infosys, is that where you would say the journey began in that manner where you saw scale incrementally?

[Speaker 1] (3:37 - 4:31)

Yeah, I think Infosys was a great learning journey in scale. You know, Infosys was founded in 1981 under the leadership of Mr. Narayan Murthy and I was one of the co-founders. You're my favorite co-founder.

I know a bunch of them. Okay, I mean. Well, anyway.

So I was part of that journey for a long time and I learned a little bit about scale. Now that's enterprise scale. Population scale is a completely different ballgame.

So enterprise scale, how to think, how to sort of look at what the future can be and sort of drive something towards that future. And then that's how I, you know, I was still, 2009 I was very much in Infosys. And then I got this offer to join the government and lead the Aadhaar project.

[Speaker 2] (4:31 - 4:36)

Infosys was the start of the IT boom in India, right? And also Bangalore became what it is today.

[Speaker 1] (4:36 - 5:17)

Yeah, I think you can think of Infosys as the original startup. Right. You know, we came, we moved to Bangalore in 1983.

From? From Bombay. Initially we were in Bombay, Pune in that belt.

And we've been there ever since. And I think, yeah, we, I mean Bangalore has had a series of technology curves, the hardware, software, internet, you know, so many things have happened. But yeah, we were part of that wave.

And I think we played a big role in establishing Bangalore as, you know, a tech city. And also I think even today I meet so many people who are entrepreneurs who said they were at Infosys.

[Speaker 2] (5:17 - 5:18)

Right.

[Speaker 1] (5:18 - 5:22)

So I think it also created a spawn of so many other people.

[Speaker 2] (5:22 - 5:42)

What worked for Infosys in a world where the incumbent large business houses were typically family businesses or, I wouldn't like to say crony capitalism, but some kind of a political element involved in a business at scale. What set Infosys apart at that point?

[Speaker 1] (5:42 - 7:20)

I think the group of founders were very synergistic. We all had different strengths. Moti was our leader.

We had common values. We wanted to build for the long term. We were marathon runners.

We were not sprinters. We built for the long term. We had a vision of creating a globally respected company of having very high standards of corporate governance.

So the whole set of values, Infosys values, which are deep in everyone. And that, I think, sort of gave the fuel to make it happen. You have to have a vision.

You have to have passion about something. You know, we had all that. And you moved from Infosys, you retired first and then Aadhaar came your way?

No, what happened was in 2009, in 2008, I wrote a book called Imagining India. So, sort of projecting how India could be in the framework of ideas. Why ideas matter and stuff like that.

And among the ideas in that book was, you know, digital ID and so on. So, at the same time, the government was planning a digital ID program. And they had got cabinet approval for that project.

And I had a chance to take up that role. So, and then my Infosys co-founder was very kind and said, yeah, you can go and do some national service. So, that's how I ended up in Delhi in July of 2009 to build India's digital program, ID program.

[Speaker 2] (7:20 - 7:24)

And before Infosys Nandan, you were in Bombay before that?

[Speaker 1] (7:25 - 8:14)

Yeah, I was born in Bangalore. In that sense, I'm an original Bangalorean. I was born in Mani Vilas Hospital in Bangalore.

And my father worked in a textile mill in Magadhi Road. Right. In Magadhi Road.

Industrial area. Yeah. And then, you know, the mill sector got into difficulties.

So, he had to leave and he thought his job would be very, wouldn't be in a fixed place. So, I got sent to Dharwad to be in my uncle's house. My uncle was very kind.

He took me in and I stayed there for five years. And then went to school in Dharwad. College also.

And then I went to IIT Bombay. And then from IIT Bombay, I joined a company which then we're all working together and then Infosys was set up.

[Speaker 2] (8:14 - 8:18)

And you met Rohini in college? I remember you guys mentioned this story.

[Speaker 1] (8:18 - 8:21)

Yeah, I met Rohini because she was in Elphinstone College. Right.

[Speaker 2] (8:22 - 8:28)

I was in IIT. Yeah. And she described you as this badass person who did something when she narrated the story.

[Speaker 1] (8:28 - 9:08)

No, no. What happened was we had a quiz competition. So, I was representing IIT in the quiz and the quiz was in Elphinstone College and obviously we won the quiz.

So, I think she was… Why obviously you won? I mean, we had that belief that we could be good at this stuff.

So, that's how I met her. That's way back in 77. She's the more outspoken of the two of you, right?

No, she's more spontaneous. She's more… I would say more emotional or mercurial or whatever.

I'm more sedate. I'm the boring type.

[Speaker 2] (9:09 - 9:12)

And she says what she's thinking right away, impulsively. Yeah, yeah.

[Speaker 1] (9:13 - 9:33)

She's obsessive about climate in particular, right? Environment is a big theme for her, part of her philanthropy. She's the biggest probably environment philanthropist in India now.

Do you agree and resonate with that belief? Oh yeah, absolutely. I mean, I'm not that involved.

But I completely agree with what she's doing.

[Speaker 2] (9:35 - 9:52)

Okay, so if we were to start the India Stack journey, I'm going to write it down because I'm learning with everybody who's going to watch this. So, Aadhaar, first thing first. So, the government asked you to build a digital ID…

[Speaker 1] (9:52 - 10:15)

The government asked me to build a unique ID. There was no digital in it. I made it digital because I said in 2009, if you're going to build an ID, it has to be a digital ID.

It has to be online, it has to be verifiable online with real time and all that stuff. So, those were the value add that my team and I did was how do we execute this using a digital first way.

[Speaker 2] (10:16 - 10:26)

So, in 2009, what was the benchmark you set for yourself while starting the Aadhaar project? Who were you looking at? Who had done well with the unique ID?

[Speaker 1] (10:26 - 10:29)

Nobody had done it. So, it was a brand new idea.

[Speaker 2] (10:29 - 10:32)

Is social security in America a similar concept?

[Speaker 1] (10:33 - 11:28)

No, social security in America came in 1936. Right. Before technology.

And in fact, the first use of mainframes was that. So, mainframes were used because there was SSN. So, SSN was a number but it was a paper-based number.

Just tell us full forms of all of this because. Yeah. So, SSN is social security number which the US did because what happened then was that Franklin Roosevelt during the 30s after the whole economic crisis in the US, they came out with a scheme where people could get benefits in their old age if they worked.

And then they had to then keep track of a person throughout the age, right? Because they would contribute while they were working and they would get their benefits when they retired. So, that's how they needed to have a number.

That's how the SSN came about.

[Speaker 2] (11:28 - 11:35)

Is that the main purpose with which governments ID, unique IDs, benefit transfer?

[Speaker 1] (11:35 - 13:44)

No, benefits transfer is a big part of it. In our case, in Indian case, there were two drivers. One was efficient benefits transfer because since 2000, India has built more and more of welfare programs which meant you send money to people.

And they had to have a solid, robust way of doing it. So, one part of this was to make benefit transfer more efficient and therefore make, you know, government targeting more efficient for benefits. The second thing was an equally important driver which was that most people didn't have IDs.

Because in many societies like in the West, 98% of births are in a hospital. And they get a birth certificate when they are born. So, that becomes the root document from which you can then get other documents.

In India, births were not happening in a hospital. They were happening at home. They were happening in a village.

So, half the babies born in some states didn't have a birth certificate. So, which was fine as long as you lived in a particular area. But if you started moving around, you know, migrating, going to the big city to enter a bus or to get a job or if the policemen stopped you on the road, you had to show some ID.

And most people didn't have an ID. So, that was a very fundamental impediment for their progress. So, inclusion of people into the formal system with an ID and making benefits transfer more efficient were the government drivers of this initiative.

And did the market play a large role in this process? No. What we, we were looking at how do you create digital infrastructure on which market or private innovation could be done.

Right. So, and we call this as public rails and private innovation. This is broadly DPI.

DPI. So, the, but, I mean, the ideas evolved, but it was actually, you know, constructed on concepts like, I'm not, on the internet. Look at the internet.

Internet was funded by the U.S. government.

[Speaker 2] (13:45 - 13:45)

Right.

[Speaker 1] (13:45 - 13:50)

But on top of that, you had all these startup companies that came up, right? Similarly, GPS.

[Speaker 2] (13:50 - 13:50)

Right.

[Speaker 1] (13:50 - 14:32)

Was again created by the U.S. Department of Defense. But on top of that, we had, you know, Uber and all these things. Mola and so on.

Maps. So, the notion that if you create a digital public infrastructure at population scale and open it up with application programming interfaces so anybody can build applications, then it'll unleash innovation. Right.

So, that was the philosophy of the design of Aadhaar and subsequently UPI. As API-led architecture, which everybody could use, and then whether it's a startup or a large company, they could then build, you know, useful applications on top of it.

[Speaker 2] (14:33 - 14:36)

Right. And when you started on the Aadhaar project, this was 2009?

[Speaker 1] (14:37 - 14:37)

Yeah.

[Speaker 2] (14:37 - 14:38)

How long did it take?

[Speaker 1] (14:39 - 15:04)

Well, I was in the government for about four and a half years. And I had made a commitment that

I would deliver 600 million people with Aadhaar by the time I quit. And I did that.

So, when I stepped down, I had, you know, 600 million people had an ID. And then, of course, it continued with the new government and now 1.3 billion people have the ID.

[Speaker 2] (15:04 - 15:10)

So, after Aadhaar came E-K-Y-C-A and E-Sign on top of it?

[Speaker 1] (15:10 - 15:14)

Yeah. No. Aadhaar was an ID that just said, Nikhil is Nikhil.

[Speaker 2] (15:14 - 15:15)

Right.

[Speaker 1] (15:15 - 15:17)

And this happened between 2009 and 2013?

[Speaker 2] (15:17 - 15:17)

Yeah.

[Speaker 1] (15:18 - 15:20)

Okay. And scaled up to 600 million.

[Speaker 2] (15:20 - 15:20)

Okay.

[Speaker 1] (15:21 - 16:13)

And with an online authentication. So, you could go somewhere, put your finger and verify that Nikhil is Nikhil. You say, my number is 123 and the system would confirm that it is that.

So, we call that as online authentication. Okay. And then everyone's ID had the name and address and date of birth and all that.

That was considered sufficient KYC to open a bank account, get a mobile connection or buy a… What does Aadhaar have? Name, date of birth, mobile number, address?

Sex, email ID and phone number. It is quite simple. But it was unique.

A person could have only one number. So, the design was how to make it unique. Right.

So, then using the additional information in Aadhaar, you could do a KYC. And then the regulators agreed that this KYC was adequate to get a bank account or a mobile connection or buy a mutual fund or whatever.

[Speaker 2] (16:14 - 16:18)

How did that work? How did you convince the regulator to accept Aadhaar as a…

[Speaker 1] (16:18 - 16:51)

Yeah. So, we had to first work with the government of India under the, you know, many acts. Because at that time, you know, prevention of money laundering act and there was lots of things happening.

So, we had to get it approved as a KYC in general. Then we had to work with each regulator. So, we had to work with the RBI to get them to accept it for banks.

We had to work with DOT and TRAI for mobile phone. IRDA for insurance. SEBI for capital markets.

So, we had to work with each regulator and we got all of them to accept it.

[Speaker 2] (16:52 - 16:53)

And how much time did that take?

[Speaker 1] (16:53 - 17:20)

That took maybe… It took several years. Two, three years.

The important thing was that the growth of that category happened at different times. For example, the use of Aadhaar KYC in banking happened after 2016. Or 2015 rather.

2015 when Prime Minister Modi wanted to do the Jandhan Yojana program. Which was to give bank accounts to everybody.

[Speaker 2] (17:20 - 17:20)

Right.

[Speaker 1] (17:20 - 17:31)

And that was a big push from the government to issue bank accounts to everyone. And Aadhaar KYC accelerated that. Because you could do a KYC immediately and get a bank account.

[Speaker 2] (17:31 - 17:38)

You mentioned somewhere that it brought down the time taken to give people bank accounts from 47 years to 9 years.

[Speaker 1] (17:39 - 17:49)

Yeah. So, on the normal GDP, if you had looked at it, it had taken 46 years to get to the level of banking penetration that we did in 9 years. Because you accelerated with digital technology.

[Speaker 2] (17:49 - 17:53)

Aadhaar KYC was the reason for that? Huh? Aadhaar KYC was the main reason for that?

[Speaker 1] (17:53 - 18:12)

Yeah. I mean, no. There was political push, government push, customer demand.

You know, banks had to execute. So, a lot of things went into it. But the technological capability of doing a KYC online, on the fly, within two minutes reduced your transaction time and cost.

[Speaker 2] (18:13 - 18:13)

Right.

[Speaker 1] (18:14 - 18:28)

So, it was an enabler. Same thing happened with mobile connections. So, we launched Aadhaar KYC in 2012-13.

But it's only when Jio got launched in 2016 that they used Aadhaar KYC to enroll a million customers a day.

[Speaker 2] (18:29 - 18:30)

Wow.

[Speaker 1] (18:30 - 18:58)

Yeah. Because they wanted to get to 100 million customers in six months. The only way to do that was to use Aadhaar KYC.

So, they used it and the rest was history, so to speak. So, they popularized the use of Aadhaar KYC in mobile industry. So, Jandan did it in banking.

Reliance Jio did it in mobile. And of course, Zerodha did it in capital markets. So, you guys used it to get…

We are big beneficiaries of Aadhaar KYC.

[Speaker 2] (18:58 - 19:00)

And did eSign happen at a similar time?

[Speaker 1] (19:00 - 19:34)

Yeah, eSign happened in 2015 when Ramsevak Sharma, who was my DG when I was the chairman of UIDA, he became the IT secretary. So, he launched eSign and DigiLocker and all these other capabilities. They came together, eSign and DigiLocker?

Yeah, they came. In 2015, they were launched at different times. Yeah, but I would say they were launched at the same time.

So, let's say DigiLocker is a wallet for government IDs? No, it's a general purpose omnibus wallet for all documents in a secure way.

[Speaker 2] (19:34 - 19:34)

Right.

[Speaker 1] (19:34 - 20:23)

Which you can keep on your phone or on the cloud. And you can keep your driver's license, your Aadhaar card, your PAN card. But it can be used by private guys too.

How did you describe that? General purpose? It's a general purpose document wallet.

Secure document wallet which can be in the phone or on the cloud. Which then you can keep all kinds of documents. You can keep your high school graduation details.

You can keep your marks card. It's not… It can be any document.

So, now there are billions of documents in this thing. More than… And what did people use eSign for?

eSign was to do digital signatures. So, you know, for example, if you buy a property and you have to sign, you can do a digital signature. Or if you're signing a contract, you can use digital signatures.

[Speaker 2] (20:23 - 20:29)

And you couldn't do this up until now because you didn't have a unique ID and you could not authorize the unique ID to sign, but now you could?

[Speaker 1] (20:29 - 20:30)

Yeah, yeah.

[Speaker 2] (20:30 - 20:38)

A number of things, yeah, but effectively that. So, eKYC came in 2012-13, eSign in 2015 and DigiLocker in 2016?

[Speaker 1] (20:39 - 20:45)

No, they both came in 2015. 2015. But the big year was 2016.

[Speaker 2] (20:46 - 20:47)

What happened in 2016?

[Speaker 1] (20:48 - 23:19)

It was a remarkable year for our digital transformation. First, on April 4th, 2016, India reached 1 billion Aadhaars. That was a landmark achievement.

April 11th, 2016, India launched UPI. Completely independent coincidence. September 5th, 2016, Reliance Jio was launched.

November 8th, 2016, demonetization happened. And December 29th, 2016, the Prime Minister launched the Bheem application. So in one year, everything changed.

Why do you add Reliance Jio to this list? Because it transformed India's mobile industry. They launched very low cost, lowest data rates in the world.

Voice was free. That created huge demand for smartphones. So smartphone prices fell.

You can see the data, they're very clear. In 2016 onwards, we saw the boom in data. Now before that, we had a voice economy.

Pre-Jio, we had what are called Minute Factories. Very cheap voice calls. But they didn't have data or their data was expensive.

What Jio did was completely collapse the price of data by creating a world class 4G network. And made voice free. So it changed the whole economics and demand perception of telecom in India.

It's a huge achievement. And what did Bheem do? Bheem essentially popularized online payments.

Because Bheem was an app launched by NPCI. Launched by the Prime Minister, built by NPCI.

And it, for the first time…

NPCI is? NPCI is the National Payment Corporation of India, which runs India's payment systems. It's an amazing organization.

So NPCI launched Bheem. And Bheem was the first exposure people got to a mobile payment app. And then of course everybody came in.

You had Google Pay, Phone Pay, Paytm, all that. And then of course today UPI does 14.4 billion transactions in a month.

[Speaker 2] (23:19 - 23:25)

Can you explain the difference between a Bheem kind of transaction and a UPI transaction?

[Speaker 1] (23:26 - 23:46)

Bheem is a particular app. So it's a consumer app on my phone on which I can initiate a payment. UPI is Unified Payment Interface, which is an underlying protocol which makes the payment happen between two apps and two different bank accounts.

Can you like break that down for me?

[Speaker 2] (23:46 - 23:54)

What do you mean by protocol? I heard a bunch of, I read a bunch of the stuff you've written as well where you use the word protocol a lot.

[Speaker 1] (23:54 - 23:54)

Yeah.

[Speaker 2] (23:54 - 23:55)

What is a protocol?

[Speaker 1] (23:56 - 24:07)

Protocol is just a way of describing the packet of exchange of a transaction. So UPI protocol is for payments.

[Speaker 2] (24:07 - 24:14)

So you're saying exchanging of information, let's say I have ABC, I want to tell you, you want to tell me DEF, that exchange is called a protocol?

[Speaker 1] (24:15 - 25:05)

No, the exchange is done through a protocol, which is a way of describing. If you look at the internet, the internet has a protocol, TCP IP, which describes how data moves on the internet. So UPI protocol is a protocol that describes how payment values move in a payment network.

And that enables dissimilar organizations to talk to each other for that category of activity. What do you mean by dissimilar organization? I mean like two banks and two mobile apps.

For example, I am on Bheem, I'm in State Bank of India. You're using PhonePe or GooglePay and you are in HDFC Bank. I can pay you in real time.

Why is that? Because the protocol, everybody talks the same. It's like there's a language of that business of payments.

[Speaker 2] (25:05 - 25:19)

Would I be right in saying if UPI is the protocol that transfers value from one store to another and its first representation is INR and the first store is a bank account, that same protocol can be used for other things?

[Speaker 1] (25:19 - 25:34)

Yeah, it can be a multi-currency protocol. It can be for wallets, bank accounts. Tomorrow it can be used for CBDC.

So once you have that basic infrastructure on that, you can add different types of accounts.

[Speaker 2] (25:34 - 25:40)

So UPI will work as seamlessly for, let's say, remittances coming into India?

[Speaker 1] (25:40 - 26:17)

No, that requires a little more work. For example, NPC of India under the ages of the RBI and Singapore under the ages of Mass, which is their central bank. They have built a connection between UPI and Singapore's FastPay or whatever it's called.

Now I can transfer money from my bank account in a Singapore bank to somebody in India and note it from Singapore dollars to Indian rupees. Yeah, but you have to build those bridges. So it's a bit cumbersome because you have to go to each country and the two countries have to come out with the rules.

[Speaker 2] (26:17 - 26:21)

So would you say that bridge is like a translation between two protocols?

[Speaker 1] (26:21 - 26:46)

Yeah, but it's also about things like, you know, fraud, prevention of money laundering, capital account convertibility, a number of issues when you do a cross-country, right? I mean, India, for example, Indians can't transfer more than $250,000 a year LRS and so on. So whereas Singaporeans have capital account convertibility.

So how do you match the different rules of the two countries? A lot of the work goes in that. Right.

[Speaker 2] (26:48 - 27:08)

So tell us more about UPI because it seems like it's this thing that changed the entire ecosystem. So you said it's a protocol, INR representation, store was one bank account to another bank account. Why did it scale in the manner that it did?

What did it provide that a traditional bank account to bank account transfer did not?

[Speaker 1] (27:09 - 29:18)

Well, at that time there was, before UPI, there was no way for two people to make a payment to each other. I mean, if I sit with you, how do I send you money? There was no way to do it.

What if I go to my bank and do, these two have the net transfer and all. Yeah, you can do net banking and all. You know, you go there, do something.

Here I can just tap on my phone and send you money. The convenience, real-time ease of use and convenience was different. Now, UPI was conceived in 2013.

And in 2015, I became an advisor to NPCI on innovation and public policy. And NPCI built it, it took three years to build it and launched it on May 11, 2016. And it was designed by NPCI, Dilip Asbe and others and Pramod on the architecture side.

Dr. Pramod Varma was also the other architect. So, all these people got together and designed this amazing, very high volume, very low cost, highly scalable payment infrastructure. So, it made sending money as easy as sending an email or sending a WhatsApp message.

Is that because now each person had a unique ID per se? No, you could have multiple. The UPI was actually more sophisticated in the sense that you could have a UPI ID.

So, it could connect to, it's called a virtual ID. It can connect to a bank account. You could use an account number.

You could use a mobile number as the address. You could use a virtual name as the address. So, Nikhil at some access bank or something.

So, all these possibilities were there. So, it just made it easy to use. And real time.

So, and then two big things happened, right? So, demonetization gave the first boost to digital payments because people had to make payments and, you know, cash was not there. And then the pandemic also, people didn't want to hand over money.

They rather point to a QR code and send money. So, both these two tailwinds also drove the, and of course the ease of use and the amazing benefits people got from it.

[Speaker 2] (29:18 - 29:53)

Right. So, DPI journey, let's say started in 2009 when you started building Aadhaar. By 2013, you had largely solved for online authentication.

Yeah, already up and running. And benefit transfer via Aadhaar in a way. Yeah.

2013, EKYC came about. 2015, eSign also happened with DigiLocker. And 2016, you had these inflection points of demonetization, Jio launch, UPI, Bheem being launched.

And Aadhaar reaching 1 billion. And Aadhaar reaching 1 billion.

[Speaker 1] (29:53 - 29:53)

At scale.

[Speaker 2] (29:54 - 30:13)

At scale. You say this a lot. Whenever you talk about a problem, you mention the word population scale a lot.

Like in every interview, I would have heard you say it 2-3 times. That's why you shouldn't watch my interviews. Why does that word matter so much to you?

Why does everything you build need to scale?

[Speaker 1] (30:14 - 31:08)

Because if you want to move the needle, and if you want to use digital technology to move the needle, you can only do it if you reach everyone. You know, when you build a private company, you build for 5-10 million people. 15 million people.

The top people, the rich people, whatever. But in building a society scale transformation, you reach everyone. And India has a billion plus people.

So you have to think, how do you do things in a way that everybody can participate? How does everybody participate? So we have to give Aadhaar to everyone.

We have to make bank accounts for everyone. Mobile connections for everyone. You know, DBT for everyone.

So the moment you want to do it for everyone, if you want inclusion to happen, you have to think at population scale.

[Speaker 2] (31:09 - 31:45)

They say when most people get very rich, very fast, they sometimes hate themselves and think the world is a terrible place and feel the world is unfair. When they get rich. When they get very rich.

Why is that? Psychology says that. I think it's a partly imposter complex, partly something.

They feel that it's unfair that they got so much whereas the rest of the world doesn't have. Okay. Post the success of Infosys, what drove you on this journey and is that why population scale matters?

[Speaker 1] (31:46 - 32:49)

No, because at Infosys, I had done a set of things. But when I got a chance to do Aadhaar, I genuinely believed that digital technology used properly can make a big difference to lives of people. So I had that belief.

And it's there in my book, 2008. So that, and then I got a chance to practice those beliefs. I mean, it's not that you can have an idea.

But somebody says, okay, fine, great idea, now go and do it. So it's a very different thing. So I got a chance to get it done and then I realized that actually we have to do a number of things.

So the last 15 years, we have built different parts of this infrastructure. And I think it's made a difference because it's made it more inclusive. Everybody is in the system.

It's helped to formalize. It's driving economic growth. All of that is happening because we have this way of thinking about digital infrastructure.

Right.

[Speaker 2] (32:51 - 33:00)

And beyond 2016, when these inflection points happened, what came in 2017, 18? When did account aggregation come about?

[Speaker 1] (33:01 - 34:38)

Interestingly, account aggregation, the first conversation also happened in 2016. Because if you look at the digital architecture, it was all about how documents could be stored, exchanged and so on. So it was really a data exchange in some way, right?

And then we wanted to implement that in different areas. And we had an architecture defined for that. And it so happened that the Reserve Bank of India had come out with the idea of account aggregator for aggregating data.

Can you explain what is account aggregation? So account aggregation is, I as a consumer, let's say I'm a financial consumer. Is it mainly for financial data?

No, it can be used for anything. But finance is the most obvious use case. So if I want to create a personal balance sheet of all my assets and liabilities, I have to get a bank statement from all my banks.

I have to get a detail of all my mutual fund assets, my insurance policies, my pension funds, my NPA, my EPFO. And bring all that together and construct my personal balance sheet. But how do you do that?

So each of them has to give that data to you. So you need again an architecture protocol to say, go to my bank, get my bank statement. Go to my mutual fund, get my mutual fund details, whatever.

So that ability for individuals to construct their balance sheet out of all the various data is what account aggregation is. But it has many uses. So one use of account...

[Speaker 2] (34:38 - 34:43)

So if I have to summarize account aggregation, should I say it's a repository of one's financial data?

[Speaker 1] (34:44 - 35:05)

No, it's not a repository. It's just an exchange. The data resides where it is.

It's a real-time system. So let's say I want a loan. I go to my account aggregator who's working on my behalf.

And there are many of them. I go to them and... And the account aggregator is not holding on to the data.

No, they don't touch the data.

[Speaker 2] (35:05 - 35:05)

Right.

[Speaker 1] (35:06 - 35:40)

They only... They're a conduit. So I go to my account aggregator and say, get my bank statement from bank X.

So it's a real-time request from me via the account aggregator of the bank. Bank gives my statement for the last six months transactions or whatever. Then I say, go to GST system and get my...

And they give you that data without being able to read the data. Yeah, they encrypt it and give it. Right.

No, they can read it because it's from the... The bank knows what the data is. But the account aggregator does not.

So when it comes to the account aggregator, it's encrypted and signed in a way that they can't peek into it.

[Speaker 2] (35:40 - 35:40)

Right.

[Speaker 1] (35:41 - 35:44)

So they have to give it as is to wherever you want to send it.

[Speaker 2] (35:44 - 35:44)

Right.

[Speaker 1] (35:44 - 35:58)

So whatever... You start from a source and give it to a destination. The source is called financial information provider.

Right. So a bank or a mutual fund. The destination is a financial information user.

[Speaker 2] (35:58 - 35:59)

Right.

[Speaker 1] (35:59 - 36:55)

The guy who uses data. So if it's a loan, as an example, I'll go to my bank, get my bank

statement. I may get two, three bank statements from two, three different banks.

I may get my income tax returns from the income tax system saying, look, I've filed my returns and paid TDS. So that data. And I'll give all this to a lender.

And then the lender looks at their data and says, yes, this guy is eligible for a loan. And all this is happening in real time, digitally. Same thing for a business.

A business can also get its bank statement, GST returns, give it to a lender and lender gives a loan. So it's actually enabling, reducing friction for financial activities. Right.

And this was conceived in 2016? Yeah. And it got implemented when?

Over the last seven years.

[Speaker 2] (36:55 - 36:55)

Right.

[Speaker 1] (36:55 - 37:33)

And it's been rolled out. It's scaling up now. There's a group called Sahamati which is orchestrating this whole thing.

And hopefully Sahamati will become an SRO, self-regulatory organization. All the banks have tied up. Many lenders are using it.

So, but also it's not just, lending is just one activity. Lending is the bulk of it today. But personal finance management, people are building apps.

Other people are building capital market apps. So it's, again, SEBI is completely on board, RBI. So it's a moment to transform financial services.

[Speaker 2] (37:34 - 37:39)

Right. What comes next in India's stack DPI journey after account aggregation?

[Speaker 1] (37:40 - 37:46)

Well, a couple of other things which happened was the Fastag system.

[Speaker 2] (37:46 - 37:46)

Mm-hmm.

[Speaker 1] (37:47 - 37:49)

Which I had designed in 2010.

[Speaker 2] (37:49 - 37:50)

Mm-hmm.

[Speaker 1] (37:50 - 38:06)

And that was to give everybody, every vehicle an ID. So just like Aadhaar gave every individual an ID, Fastag essentially gives every vehicle an ID, which is connected to a value instrument, a wallet or a bank account.

[Speaker 2] (38:07 - 38:07)

Mm-hmm.

[Speaker 1] (38:07 - 38:11)

And that streamlined the movement of vehicles on highways.

[Speaker 2] (38:11 - 38:12)

Mm-hmm.

[Speaker 1] (38:12 - 38:59)

And today that does 2.5 billion transactions a year. Will it get to the point soon where you don't have to stop? Yeah, I mean, you don't, you have to slow down.

Mm-hmm. Because it takes time for the RFID tag to be read. But it's pretty efficient, I mean, compared.

If you want to make it completely without stopping, maybe you have to use a different technology. But this is good, isn't it? It reduces from people waiting for hours to, you know, seconds.

So it's not bad. So Fastag happened a lot. Fastag was also a platform.

Fastag was a platform where a vehicle could make a digital payment on the fly. Right. So the first use case of that was electronic, I mean, tolls.

Right. It could also be used for congestion charges.

[Speaker 2] (38:59 - 38:59)

Right.

[Speaker 1] (38:59 - 39:03)

Or for parking. There are now startups that use it for parking.

[Speaker 2] (39:03 - 39:03)

Right.

[Speaker 1] (39:03 - 39:07)

So again, a platform thinking applied to vehicles.

[Speaker 2] (39:07 - 39:17)

I also wrote down IPO applications online, tax collection online, remittances, SIP plans, startups, digital IDs, digital rights.

[Speaker 1] (39:17 - 39:45)

Oh no, for example, Asba, which is when you apply for an IPO, you reserve some money for the IPO. Right. And the money is deducted from your account only if you get allocation in the IPO.

Right. That was completely made with UPI. And now half the transactions IPO are on the mobile phone, UPI, Asba transactions.

That made IPOs so much easier to do. And now they're doing IPOs for secondary markets. So I can do it.

So I think all…

[Speaker 2] (39:45 - 39:46)

IPO for secondary…

[Speaker 1] (39:46 - 40:10)

No, sorry, using UPI for secondary markets. And then the whole, you know, over time, Indian, everything goes to real time. Right.

So, you know, T plus 2 trading, T plus 1, T plus 0 trading on the capital market side, real time bank account transfers, real time, like for example, today when Indians file income tax, the refunds come in a matter of days. Right. Nowhere else you have this kind of stuff.

[Speaker 2] (40:10 - 40:34)

Right. So now me as a 20-year-old or 25-year-old, figured out the IndiaStack journey broadly. What do I build today?

That is your innovation. But give me some suggestions. I mean, look…

Let's say I'm able to raise 1 crore rupees from somebody. Nikhil, you give. And what do we build on top of any of this today, which is like low-hanging fruit, opportunities?

[Speaker 1] (40:35 - 41:57)

Well, you know, it's up to people. I mean, I look… When I see the companies that use this, I see an idea that I would never have thought of.

You know, the other day I met a company from Lucknow, which is helping roadside, you know, the street vendors to get supplies. So, they're like a B2B company that provides all the supplies. For example, if I'm running a small stall, food stall, I make omelettes or masala dosa or whatever.

Earlier, the guy… And there are many of them, right? Earlier, those guys would get up at 3 in the morning and go to the market and buy eggs and bread and, you know, spend 3-4 hours buying stuff, vegetables or whatever.

Now, these guys built an app for them to order and it's delivered to the cart. And they pay with UPI. So, this could not have been possible before that.

Suddenly, that guy saves 3 or 4 hours a day and then he can spend… He can have a longer day and he can have higher production. And that's…

Thousands of street vendors are using this. Who would have thought about this, right? Or there's like JAR, for example, which does daily savings.

You know, every day it takes a little bit of money from your account.

[Speaker 2] (41:57 - 41:59)

Do you think that's a good idea? Yeah, why not?

[Speaker 1] (41:59 - 42:02)

It was invented by Syndicate Bank, as you are aware.

[Speaker 2] (42:02 - 42:06)

There are examples of similar companies in the West which round up…

[Speaker 1] (42:06 - 42:36)

Yeah, but it's not on this… You can't… It doesn't do it on this scale.

See, in India, the pygmy deposit was invented by Syndicate Bank in the 1920s. So, they had

these pygmy deposit collectors who would go to everybody's house every day and take 5 rupees and accumulate in the bank account. So, it was invented here.

A hundred years later, we have a completely digital version of that. They do 22 transactions per month per user. That's a hell of a lot.

[Speaker 2] (42:36 - 42:36)

Right.

[Speaker 1] (42:37 - 44:23)

So, the fact is that in India, you can now build a system where I can set up an autopay and deduct 5 rupees a day from my account and save it in a bank account or save it in a gold loan or whatever they do. That is a remarkable way to bring everybody in. What is FIDE and the backend protocol?

Yeah. So, what happened was we were thinking about how to… Three of you, right?

Yeah. So, we had worked on UPI, right? So, Pramod, myself, we had all worked on UPI and we saw how a well-designed UPI protocol could unbundle payments.

And suddenly, everything changed. So, we said, if you can unbundle transactions, then you can create new business models where different people can talk to each other. Explain what you mean by unbundle.

See, let's take e-commerce, right? How does e-commerce work today? I go to some platform.

I get an app from that platform. I look up some product on that app from one of the suppliers listed on that platform who has gone through the checks of that platform. Then I order something.

And then, their delivery agent delivers it to my house. That's how. That's how integrated e-commerce is.

But, suppose you take this e-commerce transaction and unbundle it, say. And this is like UPI thinking, right? UPI also, it's not everything.

My account, your account, it's all different. So, on my app, I can place an order from any supplier listed on the ONDC grid. And I can use any delivery agent to have it delivered home.

[Speaker 2] (44:24 - 44:29)

Would you say ONDC is something built, is Beacon a protocol on which ONDC is built?

[Speaker 1] (44:29 - 44:50)

Yeah, Beacon is a protocol, which is a general purpose protocol. Explain what that means. It can be used for many applications.

It can be used for anything. It's a transaction protocol. It's a way to do transactions, to buy things.

And Fide? Fide is a non-profit which stewards this protocol. This protocol is an open source.

[Speaker 2] (44:51 - 44:54)

Just explain protocol in another way, right? So, everybody understands.

[Speaker 1] (44:54 - 45:19)

Yeah. So, it's again going back to what are the rules? What is the language of a transaction between two parties?

I need a way to describe that. So, UPI described, as I explained earlier, a payment transaction between two parties. You're talking about the rules of the language.

The actual, yeah. Actually, what are the things in it? You know, the structure of it.

What are the fields and stuff?

[Speaker 2] (45:19 - 45:22)

So, when you say Beacon is a protocol, you mean it's a bunch of rules?

[Speaker 1] (45:22 - 45:23)

Yeah.

[Speaker 2] (45:23 - 45:25)

Of the language, Beacon, in a way.

[Speaker 1] (45:25 - 46:26)

No, it's a bunch of rules for transactional stuff, for people to buy and sell things. And it can be used in commerce. So, ONDC uses Beacon as an underlying protocol.

And ONDC today does 12 billion transactions a month. It's used in mobility with Namayatri, which does 150,000 transactions a day, where people are ordering auto rickshaws. Again, you know, it's a different model than an aggregator because you pay directly to the taxi auto guy.

And it's being used in energy. We are building something called Unified Energy Interface. People can buy and sell electricity.

So, anywhere that you have to buy and sell things between different people in a consistent way, Beacon is the way to do it. Beacon is a way to do it. So, if I want to build something on top of Beacon, how do I go about it?

Well, you have to choose where you want to do it. Do you want to do it in commerce, mobility, financial services? Let's say mobility.

[Speaker 2] (46:26 - 46:34)

Let's say I want to build something using the transaction protocol Beacon, something for like transport.

[Speaker 1] (46:35 - 47:40)

This is what Namayatri has done. Namayatri has built a platform where I can find an auto on Namayatri. Exactly like I do it on my other platforms.

But when that auto comes to me, I bilaterally make a transaction and pay the auto rickshaw. And I pay 100%. And I pay immediately.

And so, this is different from being an aggregator which is sitting between you and the auto. It's just more of a discovery place. And the way the platform makes money is that the auto rickshaw driver pays a daily SaaS fee to use the platform.

So, it's a different business model. When you say he pays a daily SaaS fee, what do you mean? He pays the company 25 rupees a day to use that platform to get customers, unlimited customers for the day.

So, unlike in the other previous model, if I pay 100 rupees, then maybe 30 rupees goes to the platform in between. In this model, if I pay 100 rupees, 100 rupees goes to the driver of the auto.

[Speaker 2] (47:40 - 47:59)

So, me, if I'm a business called Namayatri or ONDC, I build on top of Beckin, this transaction protocol, a service. And then I charge a SaaS fee to the end user who uses it. Yeah.

Does that work in reducing the monopolies and duopolies which exist?

[Speaker 1] (47:59 - 49:08)

It will open up the markets and create more opportunities. I mean, so. It's the full scale of what we are saying, what can guys do?

Guys can disrupt many industries using this approach, which you could not do earlier. And you're going to start seeing that. I mean, I'm seeing some early signs.

See, the first use of these protocols was to mimic what's already happening, right? So, Namayatri already does what mobility aggregators do. Or ONDC does what e-commerce guys do.

But now, energy was a new idea. That you could now have a way to transact energy. Like charging stations.

I can go to a charging station and buy electricity through UAI. Or tomorrow, somebody may say, I'll do a Beckin-based version of quick commerce. Where instead of my investing in dark stores, I will use the retailers in my neighborhood.

Right. And I'll provide a quick commerce capability. Is that a good business idea?

I don't know. I'm just, you're asking me ideas, no? I don't know.

You're the guy who's investing.

[Speaker 2] (49:09 - 49:38)

What do you think of AI? I've heard you speak of AI in a manner where you firmly believe that large language models and building a model is going to be commoditized. No, it's already commoditized.

And nobody should spend energy on that. But you speak a lot for the Indian language and the nuance of it and building wrappers on top of these models to cater to that. Can you elaborate on that?

[Speaker 1] (49:38 - 52:20)

What do you think? Yeah. So there's a couple of, two, three different points here.

One is, large language models, in my view, will become a commodity. They are already becoming a commodity. The drivers for the Western guys to spend billions of dollars is essentially because they're in a competitive land grab situation.

So they all have to invest because not to invest is dangerous for them. So they collectively are investing $200 billion a year, buying all, you know, chips, putting data centers, all that. But they have to do it because if they don't do it, they'll fall behind.

So that's, they have a compulsion to do it. But the good news for us is all that is going to lower the cost and access to AI over time. It's already happening.

And then open source AI is coming. People like Meta have just launched Lama 3 and so, and they're going to give it away free. So I think this building LLM is a mug's game for us.

It's great for somebody there, but not for us. Our job is to leverage all this technology in a very

low-cost way to deliver value to our people. Can you give us a couple of use cases?

Yeah, so, and Indian language is the use case which we have taken as one of the top priorities because India, there are 22 official languages. There are 300 or 400, many more thousands of languages, dialects, this, that. And if you really go back to population scale, as we talked about it, how does everybody interact with the computer?

To use a keyboard, you need to know English. Even to use a touchscreen, you can do visually, but you can't read. But if you can speak to the computer, and if you can speak in your language of choice, your mother tongue, then suddenly you can access the entire knowledge in the system.

So it's a dramatic increase in access. So if a farmer in a village in, you know, UP can speak in Hindi to the computer, and the computer can go and get some information on planting techniques to him and tell him in real time back, suddenly we have made the world's knowledge accessible to that farmer in his or her language. So that's why AI is useful to bridge this gap of communication.

So that's why language AI is important for India. AI is going to be like electricity is everywhere. So everything will have some AI component in it.

We are looking at how we can use this stuff to make people's lives better.

[Speaker 2] (52:21 - 52:41)

Do you think voice will be big? I remember like when we were doing Broking 10 years ago or 15 years ago, there were systems where you could say by voice, by particular stock, X quantity, Y quantity, all of that. It never really took off.

But do you think voice controlled automation is a big use case in the near future?

[Speaker 1] (52:41 - 52:59)

Well, if you don't know how else to access, I mean, you may have a choice by either saying buy X stock for Y rupees or by keying in, buy X stock for Y rupees. We have a choice. But the guy who can't read or write, the only way is by talking, no?

Right.

[Speaker 2] (53:00 - 53:28)

So Nandan, growing up, whenever I thought of artificial intelligence, AI per se, maybe I was wrong in thinking of AI as AGI. But I thought it's when somebody else can think for me. Whichever model I've kind of used a little bit of, it doesn't feel like somebody is thinking for me.

It feels like somebody is executing better what I'm thinking. Do you think it's right to call what we have today AI?

[Speaker 1] (53:28 - 53:42)

Whatever. Those guys got together some 60 years back and called it AI. We can debate whether it's the right name or wrong name.

But I firmly believe what you said, which is the stuff is going to make our ability to do things better. Certainly not a replacement for what human beings can do.

[Speaker 2] (53:44 - 54:13)

And in terms of all this money being spent, like I track the American markets and I look at how much money these guys are spending on hardware. 200 billion across. But the revenue from AI is far lower than the capacity that people are building.

And to a large extent one can assume this capacity will become, it'll depreciate very quickly in value over the next five years. So do you think we are in bubble territory because the money going in is significantly higher than the revenue coming out?

[Speaker 1] (54:14 - 54:52)

No. At some point, if the returns don't happen and markets may choose to correct, I'm not a markets guy so I don't know about that. You decide that.

But you have to look at the incentives for these guys. They can't afford not to do it. So they'd rather spend 50 billion dollars a year each of them than not spend 50 billion dollars.

Maybe save 50 billion dollars. They make a lot of profits. But if there's a big platform shift in five years and they don't have it, then they're out of the game permanently, right?

So the cost of not doing it is higher than the cost of doing it. You have to think like that.

[Speaker 2] (54:53 - 55:00)

You're making a lot of investments in AI companies, right? I once got Tanuj in your house and I asked him, tell me all that you guys are doing. No, we do.

[Speaker 1] (55:01 - 55:08)

Most of this is philanthropic. I fund AI for Bharat at IIT Madras. This is mostly philanthropic.

[Speaker 2] (55:09 - 55:18)

But are you seeing anyone in India do anything relevant? You can tell me India and outside of India which are the companies which are doing the coolest things in AI.

[Speaker 1] (55:18 - 55:44)

Which have the biggest use case. No, I think there are many. I mean, there's a company called Sarvam which is doing AI stuff.

Bhavish is doing with Kruthim. I haven't, I don't, I'm not really interested in the company so much. My interest is how do we use this stuff at population scale, low cost, to make a difference in education or health or farming or whatever.

So what is the fint in it?

[Speaker 2] (55:45 - 55:57)

Siddharth has been talking about this for years. I think he told me first about this two years ago, three years ago. What is the fint in it and why is it so interesting as a evolution of whatever we have today?

[Speaker 1] (55:57 - 58:13)

Well, in some sense fint in it brings together the traditional, the mainstream financial system that we have and marries it to the possibilities that cryptographic techniques have taught us. You know, the concept of blockchains, the fact that everything is on one, you know, chain, immutability of data and so on. So there are some very powerful ideas that the cryptography has brought in.

What is fint in it? What is cryptography? Cryptography is, you know, the fact that you used all this technology to make things immutable.

In the sense you can, if you look at a blockchain, everybody can see the same thing. And every change made is also visible to everybody. So it gives you the tools to create, instead of having a reconciliation between all of us, that act as a public reconciliation of whatever it is, some transactions.

So that's what is used by the cryptocurrency guys. So let's separate cryptography or crypto technology from cryptocurrency. So that capability of immutability, of creating chains was used to create Bitcoin and Ether and all these things.

Basically everybody can see and verify the transactions. Yeah. Without any, in the old world, or in the current traditional world, you know, there's a custodian, there's a settlement guy, there's a regulator.

All they, the cryptocurrency guys came from this saying that we don't need authority. So when it, if you don't have authority, how do you resolve things? We'll use software to resolve it.

That's the broad theory, right? So, but there's not a good very technology there. But that has gone to its own issues because when you start building financial systems that are not part of

authority, then you end up with all the, you know, costs of that, you know, money laundering and terrorist financing and all that stuff.

And we saw what happened to many of these crypto guys. Can't that be scammed?

[Speaker 2] (58:13 - 58:36)

Like whenever I looked at the crypto world, if I were to think regulation is everybody viewing transactions verified either by proof of work or proof of stake or even proof of time. If the chain in itself is not large enough, can't it be easily scammed or conned into going in a certain direction?

[Speaker 1] (58:36 - 1:00:21)

I don't know enough to answer that. But they took a technology, but on top of that they layered currency and they layered ideology. So, they said we'll create our own currency.

And they create some 15,000, everybody creates some coins, ICOs and all that. And they said we don't want authority, we don't trust the state, so we'll do it through software. So, that got them into all their issues.

So, what we have done is we said, look, we should separate all this. There is some great technology here which we can use for high volume, low cost transactions. And if we can somehow bring that high volume, low cost transaction capability to the current world by a common way of doing it, then suddenly we can take our mainstream world and turbocharge it.

That's exactly what Finternet does. So, Finternet allows you to essentially airdrop like a modern engine into the existing system. Which is interoperable.

Which is interoperable. So, if it's in the banking system then I can take a bank deposit and following certain rules I can tokenize it. And then I can trade the tokens like very fast.

And then when I want the deposit back, I can convert it back to a deposit. So, the ability to take different kinds of assets, convert them to tokens, then use a high speed engine to transact and then un-tokenize them at the end. It's a common thing whether it's a bond or a stock or whatever.

So, that ability in a nutshell is what the Finternet does. It's an architecture to do that.

[Speaker 2] (1:00:22 - 1:00:22)

Right.

[Speaker 1] (1:00:23 - 1:00:35)

And when it's implemented, if it's implemented, what will be the first use cases of it? That depends. I mean, if it's in capital markets, it could be, you know, new types of ETFs.

[Speaker 2] (1:00:35 - 1:00:35)

Right.

[Speaker 1] (1:00:35 - 1:01:19)

Which allow you to do, which allow you to create consumer products which are different bundles of things but you can trade them very fast. Within banking it could be tokenized deposits. It could be enabling private assets to be, you know, collateralized.

I mean, we don't know exactly where this will go because we are given a framework and different sectors and different countries and different entrepreneurs and different regulators will identify different use cases. The interesting thing is, this goes back to our thinking that if you unlock something, then innovation will flourish because you have defined the rules of the game. So, that's what we are doing.

[Speaker 2] (1:01:19 - 1:01:25)

I'm going to ask you some questions like really fast answer questions. What do you think of CBDT? Where is it at?

[Speaker 1] (1:01:25 - 1:01:50)

What is the future? CBDC is a central bank, you know, sort of currency but CBDC ultimately will be one more category in the finternet. Right.

So, you need users. I think the big use of CBDC actually is wholesale. So, when you do cross-country transactions, you can do settlement much faster if you both use CBDCs.

So, then today's settlement complexities go away.

[Speaker 2] (1:01:51 - 1:02:00)

Capital account convertibility. Do you think India will change? Is it good that we are in the, like you mentioned, the LRS cap and all of that?

Will that go away?

[Speaker 1] (1:02:01 - 1:02:47)

I mean, again, it's not my decision but I believe the trend is it will go away. In my view, it will get more and more liberalized because India will get lots and lots of dollars. Already we get 108 billion dollars of remittances.

That will go to say 200 billion. We'll get FDI. And LRS outflow is maybe 20 billion.

LRS 20 billion. Inflow is, you know, 100 billion. Just remittance, forget about other things.

So, sooner or later, the system will have to liberalize. So, the way they'll do it is a little bit progressively. So, they'll take the LRS cap up or NRIs per year.

Now, they can take a million out. They'll change. So, they'll keep on the edges liberalizing it, in my view, over the next 20 years.

Whether it will be full convertibility is not my decision. It's a political decision.

[Speaker 2] (1:02:47 - 1:02:49)

But do you think more will come in than go out?

[Speaker 1] (1:02:49 - 1:03:30)

In a high-growth economy, the FDI will come, FII will come, remittances will come. So... In crypto, what's your opinion on the crypto world?

I think the crypto world has a lot of issues. And now we're seeing the cost of that. So, I think that's where the Finternet is important.

The Finternet essentially brings all the technology advantages of crypto world into the mainstream economy. Over-regulation, under-regulation. What's happening in India today?

It depends, I mean. I think some markets are well-regulated, some maybe under, some over. I mean, there's no way in the world there's one perfect answer for that.

[Speaker 2] (1:03:31 - 1:03:33)

Where do you think we're leaning on?

[Speaker 1] (1:03:34 - 1:03:53)

Which side? Some total? I think we are on par with...

I really don't know about overall regulation, but certainly on digital regulation, I think we are on par, better than most people. Socialism versus capitalism? I mean, I'm a believer in markets and capitalism.

But I also believe society must define rules for competition and innovation.

[Speaker 2] (1:03:55 - 1:04:02)

And last words to a 20-year-old me, starting work tomorrow with one crore that I borrowed from someone. What do I do?

[Speaker 1] (1:04:03 - 1:04:44)

I think, I've never seen so much opportunity. I think the ability for young people today to create successful companies has never been higher. But I'm confused with too much opportunity.

I want you to tell me one... No, that is... You have to, I mean, you have to...

There is something called the tyranny of choice. You know, there's so many things to choose from. You go to a supermarket, there's 17 varieties of, you know, cheese or something.

Then you get lost. So, half of life is deciding what not to do. So, you chip away and remove stuff from your thing and focus on a few things.

But the opportunity is huge.

[Speaker 2] (1:04:45 - 1:05:00)

Sector-wise, any particular opinion? Like, right now we're hearing a lot of defense is big, energy transition is big. Like, I personally have been researching what to do in energy transition for a year.

Any outlook on that sector or any insight that...

[Speaker 1] (1:05:00 - 1:05:34)

No, energy transition is huge. It's going to involve trillions of dollars. But where do investors make money in the private markets is not clear.

Because energy transition is a big capital gain, right? I mean, building massive solar plants or stuff like that. So, we have to figure out...

See, there's two, there's big energy and little energy. Big energy is the grid. Little energy is thousands of homes having batteries and rooftop solar.

So, the opportunity is actually in this little energy where there's lots of players. And how do you make that efficient? How do you create markets there?

Right.

[Speaker 2] (1:05:35 - 1:05:40)

Yeah, that's it. Thank you, Nandan. Thank you so much for taking the time to do this.

Thanks, Nikhil. I hope you didn't get bored.

[Speaker 1] (1:05:40 - 1:05:49)

Not at all. Very, very exciting. Okay, super.

Thanks, guys. That's it. I'm done.

I want to go.

[Speaker 2] (1:05:51 - 1:06:01)

How long was it? 1 hour and 15 minutes. Okay.

Why are you still shooting for him? Yeah. He's been shooting for 7 years.

I know.